

MOHAMMED NASEEF

AI/ML Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

B.Tech Computer Science graduate with hands-on experience in machine learning, deep learning, and GenAI. Trained and deployed CNN and transfer learning models achieving up to 97-100% accuracy. Developed RAG pipelines using LangChain and ChromaDB, and deployed REST APIs using FastAPI. Proficient in Python, Pandas, NumPy, SQL, and Git; experienced working in Jupyter and VS Code environments.

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, OOP), SQL, C

Machine Learning: Supervised Learning, Unsupervised Learning, Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, CNN, Transfer Learning, Neural Networks, Feature Engineering

Generative AI: LangChain, LangGraph, RAG, Prompt Engineering, HuggingFace Transformers, Vector Databases, Embeddings, FAISS, ChromaDB, Google Gemini API

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, EDA, Power BI, Tableau, Excel

Computer Vision: OpenCV, Image Classification, Image Processing, Object Detection

Development Tools: Jupyter Notebook, VS Code, Git, GitHub, FastAPI, Streamlit, Docker, Docker Compose, GitHub Actions, Linux, REST APIs

EXPERIENCE

AI/ML Engineering Intern

Aug 2025 – Present

Catalyst Skill Hub, Calicut, India

- Developed end-to-end AI/ML data pipeline for predictive engine failure detection across 3+ sensor datasets; performed data ingestion, cleaning, and feature engineering using Pandas and NumPy in a reproducible Jupyter Notebook environment.
- Designed leakage-safe ML pipeline using Scikit-learn Pipeline and ColumnTransformer to apply transformations consistently across train and test sets; achieved ROC-AUC 0.81 with 92% recall on ride cancellation prediction.
- Conducted EDA using Matplotlib and Seaborn across 5+ real-world datasets to validate data quality, identify distributions, and surface anomalies; documented findings to support collaborative AI model development.

PROJECTS

AI Mentor — Personalized AI Learning Platform

Python, FastAPI, React, PostgreSQL, ChromaDB, Google Gemini API

- Designed a full-stack GenAI-powered learning platform with a Prompt Engine that generates AI personas from user onboarding data stored in PostgreSQL, covering 4+ learning domains.
- Engineered a RAG pipeline from scratch using ChromaDB and Gemini Embeddings to ground every AI response in a curated knowledge base; developed an Agentic Curriculum Generator producing structured JSON roadmaps rendered as interactive React UI.
- Implemented JWT authentication, async FastAPI backend, and 3 independent AI microservices for modular deployment.

Repository: github.com/Naseefnf/ai-mentor

MachineSense — AI-Powered Factory Machine Identification Platform

Python, FastAPI, TensorFlow, ResNet50, React, SQLite, Docker, GitHub Actions

- Developed a full-stack multi-tenant SaaS platform using React + FastAPI with isolated data and AI models per company; containerized with Docker Compose for one-command deployment and CI/CD via GitHub Actions.
- Implemented Transfer Learning with ResNet50 achieving 97-100% validation accuracy with as few as 80 images per class using data augmentation and Dropout (0.5) regularization.
- Constructed no-code AI training pipeline enabling non-technical managers to trigger ResNet50 fine-tuning with one click; engineered 10+ RESTful API endpoints with JWT role-based authentication reducing onboarding time by 60%.

Repository: github.com/Naseefnf/machinesense

Waste Classification System

Python, TensorFlow, CNN, OpenCV, Arduino

- Trained CNN model on 2,000+ labeled images to classify biodegradable vs. non-biodegradable waste; achieved 91% accuracy and 0.89 F1-score with real-time OpenCV inference under 1.5s latency.
- Integrated ML model with Arduino-controlled servo for automated physical waste segregation at 95% sorting success rate across 6 waste categories.

Repository: github.com/Naseefnf/WastepyWasteSortingUsingMLandIoT

EDUCATION

MES College of Engineering, Kuttippuram

2021 – 2025

B.Tech in Computer Science and Engineering

COURSES AND CERTIFICATIONS

Artificial Intelligence and Machine Learning — ICT Academy, Cyber Park, Calicut

Python for Data Science and AI — IBM, Coursera (2024)

Prompt Engineering for Everyone — IBM, Coursera